LESSON PLAN

GAS TURBINE ENGINE (16)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Magnetic particle inspection application advantages, limitations, circulars and longitudinal magnetization, general procedure for magnetic particle inspection
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Magnetic particle inspection application advantages, limitations, circulars and longitudinal magnetization, general procedure for magnetic particle inspection various types of parts which can be inspected after inspection reasons for demagnetization
5. Specific Objectives	16.1.1.1 Trainee will be able to describe types of NDT and perform liquid penetrant, magnetic particles and eddy current inspections

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Circular and Longitudinal magnetization using Yoke and Coil methods; Safety precautions regarding electrical shock and eye protection; Instructor supervision of field application on test blocks
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) List three types of Non-Destructive Testing methods.
(b) Why is Magnetic Particle Inspection preferred for ferromagnetic materials?
(c) What is the primary difference between Circular and Longitudinal magnetization?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Application advantages and limitations of MPI • Types of parts suitable for magnetic particle inspection • Reasons for demagnetization after inspection	L & Ds	PPT, IB	15 min
(b)	• Principles of Circular magnetization techniques • Principles of Longitudinal magnetization techniques • Selection criteria for magnetization methods	L & Ds	PPT, PDF	20 min
(c)	• General procedure for magnetic particle inspection • Interpretation of indications and false calls • Post-inspection demagnetization procedures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain why demagnetization is necessary after inspecting a component.
(b) Differentiate between the indications produced by circular and longitudinal magnetization.
(c) Describe the general steps involved in the magnetic particle inspection procedure.

12. Home Assignment and Follow Up	Prepare a brief summary comparing the advantages and limitations of Magnetic Particle Inspection with Liquid Penetrant Inspection, referencing the previous lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Eddy current inspection, principle of operation, function of basic system, coil impedance, electrical conductivity mag permeability, types of inspection coils, application of various types of parts	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min